DLS Report

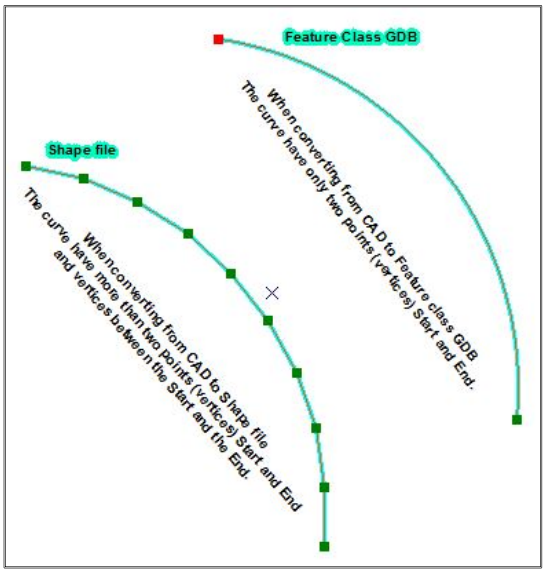
summary report to help our technical support team to understand certain problems That DLS workers face in their daily works, these problems include problems in their Data and problems in how to facilitate the work process by generating alternative solutions.

- some of these problems:
  - Problem 1: How to solve the problem of vertices through the curves in (SDE) data?
  - Problem 2: How to get page layout for parcel corners and its coordinates listed in a table?
  - Problem 3: How numbering Parcels (Temp_ID) automatically?
  - Problem 4: Some Extensions and toolbars and Tools they need to facilitate the work?

After the discussions with "E.Awad alawadi" about how to complete works using ArcGis instead of AutoCad by finding the alternative tools that can replace AutoCad tools, we decided :

- to have a good idea about DLS WORKS, we start working with each section to know what each one need to complete his works properly. These sections:
  - Treatment Section قسم المعالج ...I did not see this section.
  - Transaction section قسم المعاملات...Using AutoCad products, I saw this section. I got help from (E.Ayat Qaisiya and E.Imad Tarifi) .
  - قسم الإدامة ...using ArcGis Product, I saw this section.
    I got helf from (E.Shireen and E.Sarah).
  - Audit Section قسم التدقيق ...Using AutoCad and ArcGis products, I saw this section. I got help from (E.Ahmad thenibat).

- **problems-solutions :**
  - **Problem 1: How to solve the problem of vertices through the curves in (SDE) data?**
    1. Each curve should have two vertices, the **Start** and the **End** vertex.
    2. The problem created when they export CAD files to SDE, they export the CAD files to Shape files, so all curves drawn as a series of vertices.
    3. But if they export CAD files TO a Feature class geodatabase the curves will be drawn as Start and END vertices.

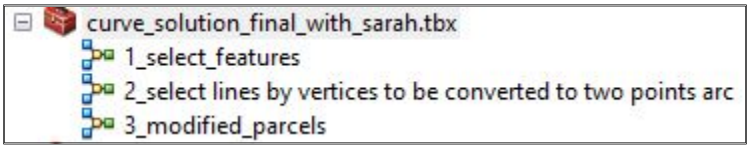


4. The solution of this problem:
   Requirements:

   1) Add “Curves and lines add-ins”

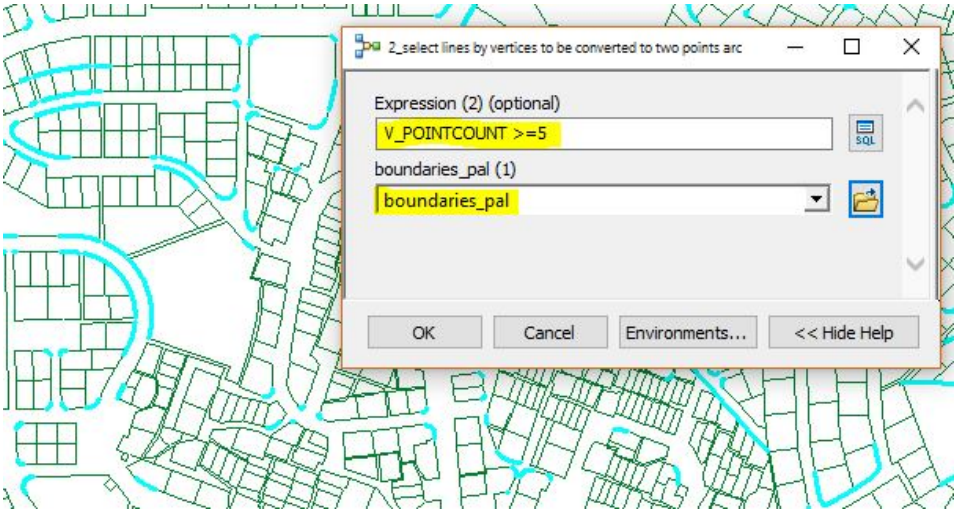


   2) Follow the steps one by one using modelbuilder tools :

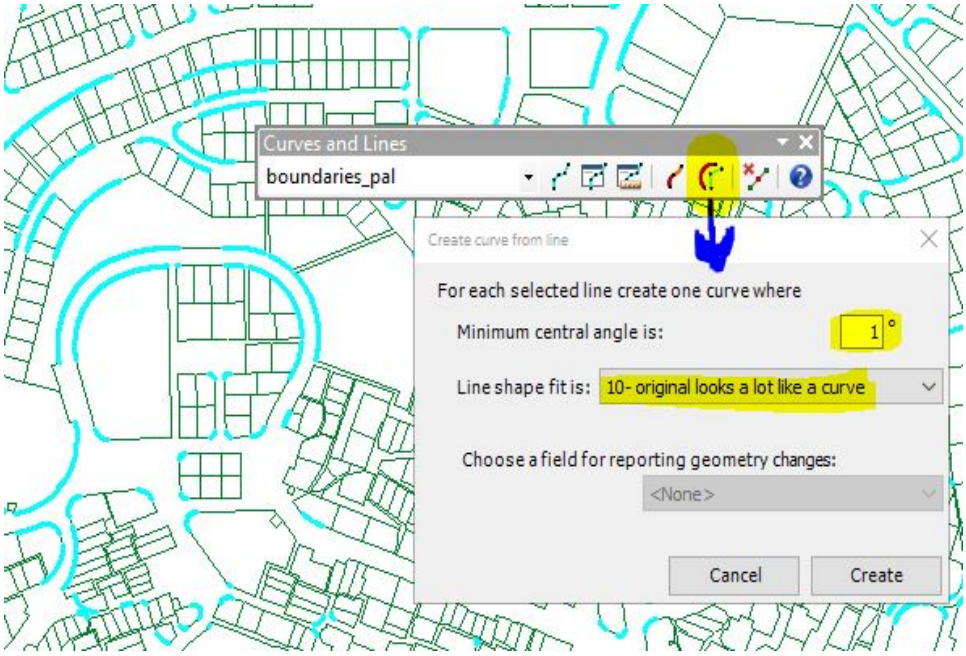


   - First tool: “1_select_features”, this tool extract three layers (parcels, boundaries, corners) within specific area (hods layer).

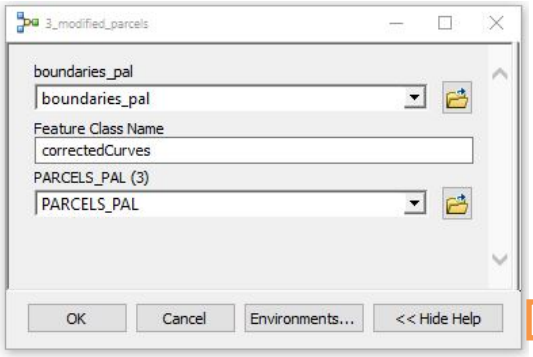
- Second tool: ” 2_select lines by vertices to be converted to two points arc”

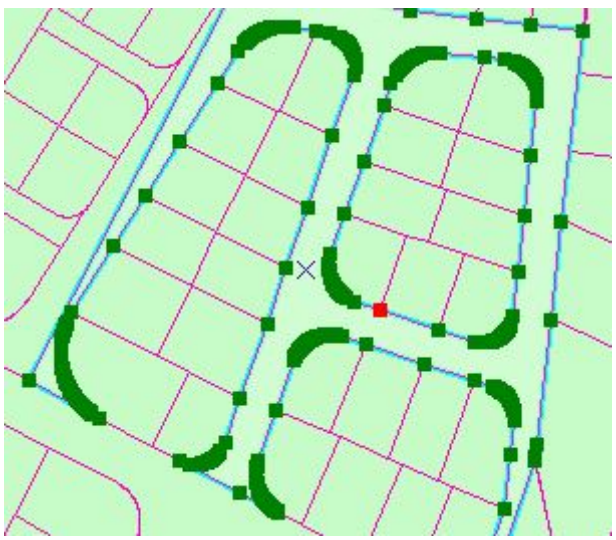


- Step 4: Use the “Curve and lines add-ins” to convert the selected lines and arcs which have more than two vertices to arcs with two vertices, start and end.

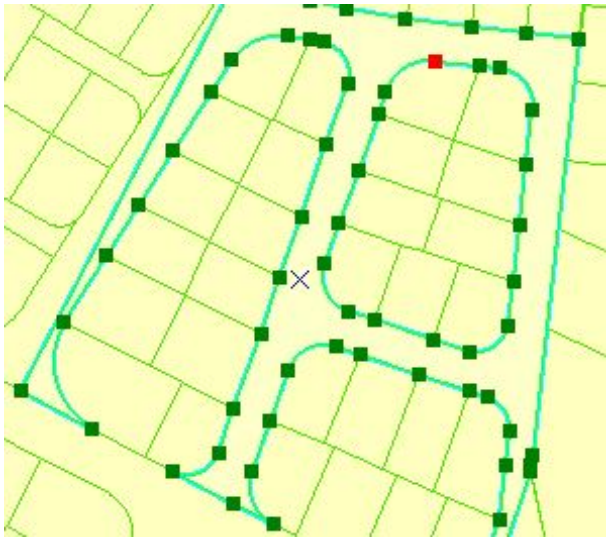


- Third tool: “3_modify_parcels”…this tool fix the arcs in the (parcels_pal) feature class based on the boundaries which just fixed in step 4 and preserve all the attributes of the (parcel_pal).
  After finishing running this tool, the boundaries of the (parcels_pal) polygon feature class will be the same as the (boundaries_pal) line feature class.





Parcels arcs before processing



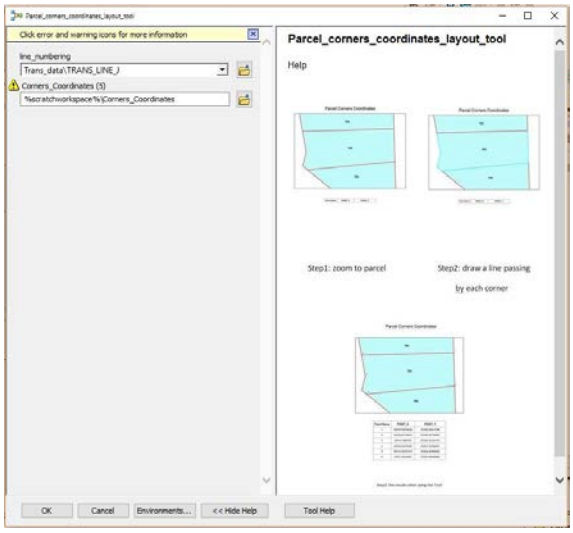
Parcels arcs after processing

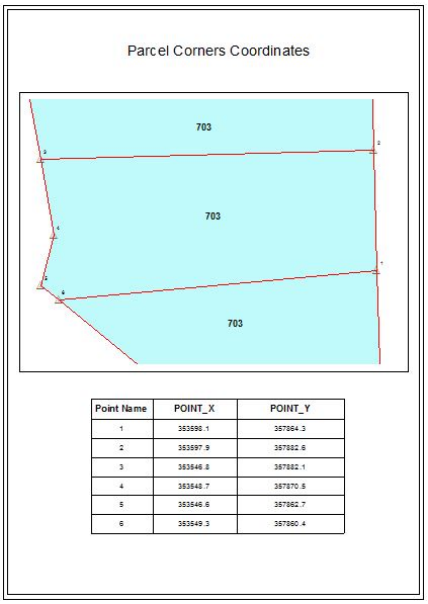
- **Problem 2: How to get page layout for parcel corners and its coordinates listed in a table?**

Prerequisite: Mapping and Charting Solution license…

Production mapping extension

*We need this extension to use ***Graphic Table Element*** in the layout page help us updating the coordinates each time you change the parcels corners.





*for more details click the **attached file**

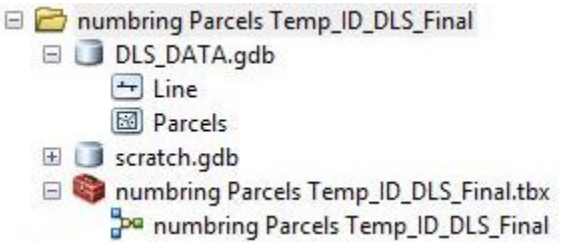
Corners_Numbering with Coordinates for

- **Problem 3: How numbering Parcels (Temp_ID) automatically?**

  To perform this task…we make a modelbuilder tool to automatically numbering the **Parcels (Temp_ID)…**

  ***The Tool Help:**

  <u>folder location and subfolders</u> :



- Scratchworkspace.gdb: the default and scratch file Geodatabase in which ***the intermediate feature classes*** will be saved.

-numbering Parcels Temp_ID_DLS_Final.tbx

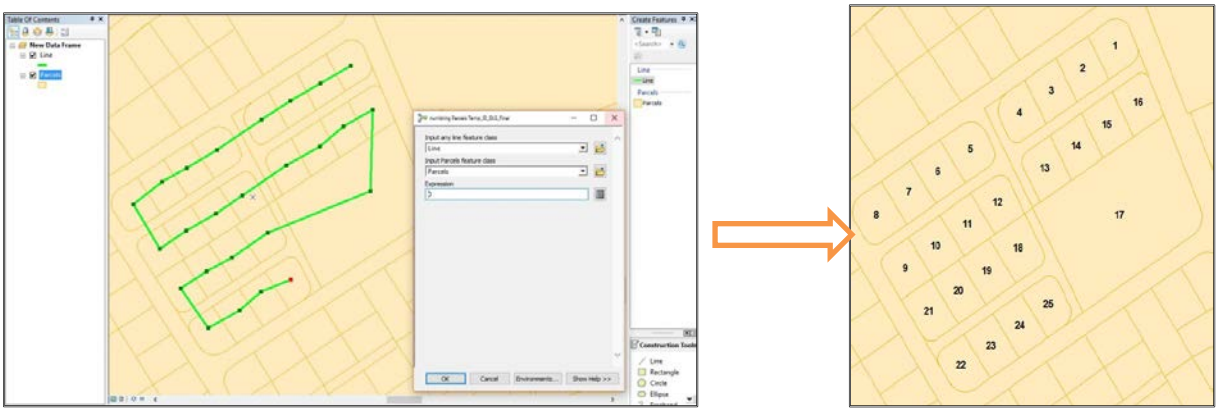
- DLS_DATA:

  1-**Parcels** Feature class: the numbering will be in **TEMP_P_ID** field

  2-**Line** Feature class: the line used to pick vertices in each parcel we want to number it.

<u>Prerequisites:-</u>

1- Start Edit with any line feature class.
2- Zoom to the parcels which you want to number their **TEMP_P_ID** .
3- Draw a line and click inside each parcel you want to number it.
4- Use the tool by dragging the **line** feature class and the **Parcels** feature class and the **starting_number**.



<u>These figures show the parcels before using the tool and the results after running the tool.</u>

*for more details click the **attached file**

numbering Parcels Temp_ID_DLS_ Tool H

- **Problem 4: Some Extensions and toolbars and Tools they need to facilitate the work?**
  - **Mapping and Charting Solution license >>> production mapping extension**
    - **Production Editing Advanced toolbar:**



  - **License type: Advanced**
    - **COGO Toolbar:**



- **COGO Report button**  **:**

Icon	COGO description	What it reports
	Direction and distance of a line	Reports the direction and distance between two points you click on the map.
	Angle between two lines	Reports the acute angle between three points you click on the map.
	Orthogonal distance between a line and a point	Reports the orthogonal distance and direction between two points you click on the map and a third point.
	Area between points	Reports the area and perimeter of a series of the points you click on the map. The boundaries of the area are always straight lines.
	Coordinates of an existing feature	Reports the coordinates of a point or line feature you click on the map. If the feature is a line, it needs to be a COGO line, either entirely straight or entirely circular.
	COGO description of an existing line feature	Reports the COGO description of a line you click on the map. The feature needs to be a COGO line, either entirely straight or entirely circular.
	Orthogonal distance between an existing line feature and a point	Reports the orthogonal distance and direction between a line feature you click on the map and a point you click on the map. The feature needs to be a COGO line, either entirely straight or entirely circular.

Items available in a COGO report

- **Proportion button**  **:**
  Splits the selected line into a number of segments.

  *note: Any difference between the line length and the sum of the length of the segments is allocated proportionately to the new line's length.

    - **Advanced Editing:**

